Emanuel Camacho

MSc Aerospace Engineering Candidate • Aerodynamics & Propulsion • Control Systems
emanuel.aerospace@gmail.com | +351 960 366 633
linkedin.com/in/emanuel-camacho-05ab8a245 | emanuelcamaerospace.github.io/website

EDUCATION

Instituto Superior Técnico — Expected Nov 2026
MSc in Aerospace Engineering — *Lisbon, Portugal*
- Specialisation in Aerodynamics and Aircraft Propulsion Systems.

Technical University of Munich (TUM) — Aug 2025
Erasmus Exchange Semester, MSc in Aerospace Engineering — *Munich, Germany*
- Coursework in aircraft propulsion and aircraft system integration.

Instituto Superior Técnico — Jul 2024
BSc in Aerospace Engineering — *Lisbon, Portugal*
- Relevant coursework: Fluid Dynamics, Flight Mechanics, Flight Control.

RESEARCH

Instituto Superior Técnico & Portuguese Air Force — Sep 2025 – Nov 2026
MSc Thesis: Laser-Based Interception of Unmanned Aerial Systems — *Lisbon, Portugal*
- **Defence-funded research:** Developing a low-power counter-UAS (C-UAS) prototype that neutralises drone optical sensors, funded by and in collaboration with the Portuguese Air Force (*Força Aérea Portuguesa*) and the Institute for Systems and Robotics (ISR).
- **Control and targeting:** Designed and implemented an Integral Sliding Mode Controller (ISMC), tuned with a genetic algorithm, for a precision 2-axis gimbal, enabling dynamic visual tracking of moving targets.
- **System simulation:** Built an end-to-end Simulink framework combining high-fidelity gimbal dynamics, a visual detection pipeline, sensor noise models and target trajectory physics to validate the system before hardware deployment.
- **Hardware integration:** Prototyped a lightweight C-UAS platform with custom 3D-printed mechanics, Pixhawk-based embedded control, servo actuators and a dedicated power-regulation architecture.
- **Experimental validation:** Ran field tests to evaluate optical tracking and neutralisation performance, establishing baseline metrics for mobile and drone-interceptor deployments.

EXPERIENCE

AeroTéc – BLUE Turbojet Project — Jan 2026 – Present
Design Team Member, Compressor Section — *Lisbon, Portugal*
- Carry out the aerodynamic design and optimisation of the centrifugal impeller and diffuser for a small-scale turbojet engine, as part of a 15-member team.
- Developed MATLAB thermodynamic and mean-line flow scripts to compute 3D blade geometry, flow paths and stage pressure ratios.
- Built the team's automated parametric CAD framework by linking MATLAB to Siemens NX, reducing each design iteration to a single click and around 30 seconds of processing.
- Run preliminary 3D CFD simulations in Siemens STAR-CCM+ to assess boundary-layer separation, optimise blade incidence angles and verify predicted pressure ratios.

ANAC – Portuguese National Civil Aviation Authority — Aug 2024 – Sep 2024
Project Intern — *Lisbon, Portugal*
- Developed models predicting the post-failure descent of commercial UAS, computing ballistic trajectories and ground-impact zones after critical system failures.
- Built an interactive browser-based simulation tool that allows certification engineers to assess UAS safety parameters.

SKILLS & LANGUAGES

Software: Siemens NX, SolidWorks, Solid Edge (CAD); Siemens STAR-CCM+ (CFD); MATLAB, Simulink, Simscape; Python, C; LaTeX.
Engineering: Turbomachinery Design, Aerodynamic Analysis, Computational Fluid Dynamics, Nonlinear Control (Sliding Mode), C-UAS System Architecture, Systems Engineering.
Languages: Portuguese (Native), English (C2), Spanish (C1), German (A2).

CERTIFICATIONS

STAR-CCM+ CFD Training Certification – Cadflow / Siemens Digital Industries Software — Sep 2026
German Language, Level A2 – Volkshochschule Lernportal — Aug 2026
German Language, Level A1.1 – Technical University of Munich — Aug 2025